KCM Heat-Shock Transformation

PROTOCOL

1. Turn on the **EchoTherm**: **block A 4 °C**, **block B 42 °C**. Wait until both are at temperature and leave them there.
2. Warm **Carb** plates to room temperature and **label the bottom**: date, your initials, strain, plasmid, selection. Cold plates are hard to write on.
3. Put a **100 µL** competent cell aliquot on block A and thaw (~30 s). Add **25 µL KCM** and pipette gently to mix. One aliquot does about **3 reactions**.
4. Put the DNA tube on block A and let it cool **30 s**.
5. Add **40 µL** of the cell/KCM mix to the DNA. Mix gently.
   – 40 µL to **10 µL DNA** is the standard — it assumes the DNA is about 20% of the total.
   – A large reaction (~20 µL): use 100 µL, or the whole tube of cells.
   – Retransforming a miniprep: **0.5 µL** plasmid into 10 µL cells.
   – Too much DNA dilutes the salts and **drops the efficiency**.
6. Hold on block A: **4 °C for 10 min**.
7. Move to block B: **42 °C for 90 s**.
8. Back to block A: **4 °C for 1 min**.
9. Plate everything on **Carb**. Incubate **inverted** at 37 °C overnight. **No rescue needed** for Amp/Carb.
10. Cancel the temperature programs when you are done.